SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

Duration : 45 Minutes
Total Marks:20

QUESTIONS: 2

Annexure A

Descriptive Written Test

Code of Conduct:

I Pawan Kumar C | 192514(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Pawan.C

Level-4

Pavan Kumar . C
192511446
CSA1714
AI
25/07/2026.

C01-AT.3-Descriptive Test.

1. Intelligent Agents in a smart Home automation system.

A smart home automation system uses intelligent agent in to monitor the environment, make decisions, and perform actions automatically. It receives inputs from sensors such as motion detectors, temperature sensors, light sensor, security cameras and time schedules to control devices like light, air conditioners, heaters. and security alarms.

1. Simple Reflex Agent:

A simple reflex agent acts only on the current input without considering past events.

How it works:-

* If motion is ~~get~~ detected in a room, turn on the lights
* If no motion is detected for 10 minutes, turn off the lights.
* If smoke is detected, activate the fire alarm.

Advantages:-

* Fast response
* Easy to implement.
* Suitable for simple tasks.

Disadvantages:

* cannot rember previous States.
* Cannot adapt to user preferences.
* Poor performance in complex situations.

2. Model-Based Reflex Agent.

A model-Based Agent maintains an internal model of the environment and uses past information along with current inputs.

How it works:

* Remembers which rooms are occupied.
* Tracks indoor temperature trends.
* Keeps the status of door and windows.
* Decides whether heating or cooling is needing based on both current and previous conditions.

Advantages:

* works well in partially observable environments.
* More intelligent than simple reflex agents.
* Makes better decision using memory.

DisAdvantages:

* More complex to design.
* Require additional memory and processing.

3. Goal-Based Agent.

A Goal Based Agent selects actions that help achieve predefined goals.

Example goals:-

* Maintain room temperature between 22°C and 24°C.
* Ensure how security when occupants are away.
* Reduce.

How it works:

* If everyone leaves the house, lock all doors, switch off light, and activate security mode.

* plans the best sequence of actions to achieve comfort and safety.

Advantages:

* Makes planed decision.
* Flexible for different goals.
* can evaluate multiple action choices.

Disadvantages:

* Requires more computation.
* Slower than reflex agent.

4. Utility - Based Agent

A Utility - Based Agent choose the action that Provides the highest over all benefit by balancing multiple objective.

How it works:

The agent consides:

* user comfort.
* Energy efficiency.
* security.
* Electricity cost.
* user preference

Example:

* Increase AC temperature slightly to save energy.
* Dely charging smart applications.
* Maintain acceptable comfort while reducing Power consumption.

Advantages:

* Produces optimal decisions.
* Balances conflicting objectives.
* Learns and adapts over time.

Disadvantages:

* Most complex to implement.
* Requires more processing power and data.

Conclusion:

A hybrid intelligent agent provides the best performance for a smart home because it combines quick reactions, environmentals awareness, goal-oriented planning and utility optimization. This approach ensures:

* Comfort: Learns and maintains user preferences for lighting and temperature.

* Energy Efficiency:- Reduces unncessary power usage and optimizes appliance operation.

* Security:- Continuously monitors the home, detects threats and responds appropriately.

2. Robot in a Unknown Maze

uniform cost search (UCS).

* Expands the node with the lowest path cost

* Finds the optimal (lowest-cost) path.

Advantage:

It have a complete and optimal.

Disadvantages:

High time and space complexity (Time: $O(b^{1+[C^*/\epsilon]})$, Space: same order.

Iterative Deeping search (IDS).

* Repeatedly performs depth-limited search with increasing path.

* Low memory (space) complete for finite depth.

* Take more time not optimal when cost differ.

Advantages :-

Low memory (space : $O(bd)$), complete for finite depth.

Dis advantages :-

Repeats searches, so take more time (Time $O(b^d)$); not optimal when cost differ.

Intelligent Agents :-

* Simple Reflex :-

reacts only to current state ; poor ~~of~~ for unknown mazes.

* Model Based :-

~~plans actions to reach the exit.~~
Remembers explored paths and obstacles.

* Goal Based :-

plans actions to reach the exit.

* **Utility Based:-**

choose the best path by considering cost, time and safety.

Best Choice:-

A Goal-Based + model Based agent with uniform cost search (UCS) is the most effective because it remembers explored areas, plans towards ~~the~~ exit and finds the lowest cost path.